PD Paper 3 SAMPLES

SAMPLE C

Question 1

One issue in the interpretation of the graph is how computers (screen time) influence mental health. Source 1 shows that an unhealthy work-life balance results in anxiety and depression. An unhealthy work-life balance could be when a student puts too much time into school work, for example, staying up late and never giving themselves a little break to get distracted. Such continuous exposures to the stresses and monotony of school work lead to symptoms of depression. Depression is closely linked with anxiety. Perhaps the reason that students may stay up until late and work on their computers is that they are anxious they won't do so well and will get criticised by teachers and parents. This anxiety can keep them up at night and contribute to depressive thoughts and feelings.

Excessive screen time is unhealthy and should be avoided as best as we can. It is not possible to avoid it entirely in today's educational context, of course, because it becomes more and more dependent on technology. This was especially evident during the COVID pandemic when students had to spend a considerable portion of each day in front of the computer, which has been acknowledged as a major reason for the increased prevalence of mental disorders during that period (and still).

Question 2

Source 2 was designed as a true experiment: the independent variable was the amount of time spent on screen on the day of the study. It was operationalised as the task for the students: either an online task that would take up to 6 hours or being asked to stay off the screen for the entire day. The dependent variable was the score on a Mental Wellness Exam taken at the end of the day.

As this was a true experiment, we can interpret the findings as the extent to which screen time "influences" wellbeing. However, one needs to take into account potential biases that may be introduced by confounding variables. For example, no screen time may have resulted in less work done and more communication with friends.

Results of descriptive statistics suggest that the scores on the mental wellbeing exam were lower in the on-screen group (although no definitive conclusions can be drawn before an inferential test is conducted). Scores in the on-screen group range from 7 to 21 with the mean 11.7 and the median 10. Scores in the off-screen group range from 12 to 24 with the mean 17.64 and the median 15. This suggests that all parameters in the off-screen group are shifted towards the higher end.

Standard deviation seems to be a little larger in the off-screen group (4.48 versus 3.87). Standard deviation is a measure of how much individual results are spread around the mean, so one might say that there is a larger variation in mental wellbeing among individuals who stayed off the screen.

Looking at the box-and-whisker plot, we can see two patterns of distribution that are very similar to each other. For example, in the on-screen group 50% of individuals scored between 7 and 10. Within that, 25 % of the sample scored either 9 or 10. So the distribution probably has a spike on the left side. After this there is a larger spread of values and a more gradual slope between 10 and 21. 50% of individuals scored between 10 and 14 and another 50% between 14 and 21. This pattern of results is very similar in the other group. Nevertheless, the spread between Q1 and Q3 is somewhat larger, echoing what we saw in standard deviations.

Overall results suggest, descriptively, that the distribution of scores in the off-screen group is shifted to the right. An inferential test might show a statistically significant difference between groups.

Based on the box-and-whisker plot, it appears that the distribution of results in both groups is not normal (skewed to the left). This suggests that the mean and the standard deviation may be inappropriate as descriptive statistics in this particular case.

Question 3

Focus group studies can be heavily influenced by researcher variables, for example, when the researcher conducting the focus group consciously or unconsciously steers the conversation in a direction that is more aligned with his / her own expectations or biases. Additionally, focus group results may be affected by nuances of interpersonal dynamics in the group. For example, there could be dominant participants who take over the conversation and impose their own agenda, making others in the group reluctant to speak their mind.

Credibility refers to the extent to which results of the study can be trusted, or the extent to which the study conclusion reflects the real state of things in the world. Credibility of a focus group is difficult to maintain due to the reasons mentioned above.

To improve credibility of her findings, the researcher could use reflexivity checks. This is when at various stages of the analysis the researcher stops and asks herself, could my personal expectations and biases have affected my interpretation of this part of the transcript? Everybody has a personal history and it is reasonable to expect that it will affect the researcher’s perceptions at least in some ways. Such influences are impossible to eliminate, so the best we could do is acknowledge them.

Another way in which credibility could be improved is by staying as close as possible to the text of the transcript. Results of a focus group are recorded word-for-word. When conducting an inductive content analysis, the researcher then goes through the transcript and identifies "emerging themes". In doing so, there is always a choice - to stay close to the text and use participants' own words to describe their experiences, or introduce a bit of interpretation and take the liberty to summarise what participants said in the researcher's own words. Staying close to the text and gradually combining themes into larger and larger categories could improve the credibility of findings and reproducibility of results.

In conclusion, there are a few things that could be done to improve credibility of this.study, in particular, reflexivity checks and staying close to the text when conducting inductive content analysis.

Question 4

The evidence presented in the exam material as well as my background knowledge of the subject suggests that the use of technology can influence the mental health of students to some extent.

A recently conducted study with a large cross-cultural sample of participants provided correlational evidence that participants who spend excessive amounts of time on electronic devices also tend to be more likely to develop symptoms of mental disorders such as depression, anxiety or OCD. Perhaps this is not due to the use of a gadget as such, but due to the decreased social contact and other factors associated with it.

When using social media, people may get exposed to unrealistic images portraying happy and successful life and this could contribute to a sense of worthlessness and inferiority, resulting in depressive tendencies. When playing video games, they may become addicted and lose out on other developmental opportunities, falling behind in their school work and further reinforcing lack of self-worth. There are various pathways through which the use of technology may result in negative consequences for mental health.

We have to acknowledge though that technology does not always have a negative effect and can even have positive effects sometimes. An example is using virtual reality in treating PTSD. By recreating traumatic events in a highly controlled virtual space and gradually training patients to be exposed to adverse stimuli without triggering a PTSD response we can help patients cope with the problem.

Evidence provided in the research supports the view that the use of technology negatively impacts mental health.

For example, the researcher in source 3 conducted a focus group study among students who had to go through online learning during COVID and discovered among the emerging themes such themes as feelings of missing out, loneliness and anxiety. Focus group falls under qualitative research, so the conclusions may not be generalisable to the entire population, but the conditions in which these students found themselves were quite similar for all students around the world. In a qualitative study it is also not guaranteed that another researcher would arrive at similar conclusions, so researchers may consider triangulating their conclusions with other data.

In Source 4 it was discovered that hours spent doing homework on laptops correlates positively with a number of measures of mental health problems, namely, cognitive dulling, depression, general anxiety and OCD. The correlation with depression and general anxiety was the strongest (0.76 and 0.80). These correlation coefficients suggest a very strong relationship. We should remember of course that correlation does not imply causation and that mental health can also influence how much time students choose to spend on their laptops. On the other hand, the amount of time spent doing homework (be it on a laptop or not) can be an indicator of general struggles with school performance. Perhaps it is not screen time as such that was influencing participants.

Finally, source 5 also clearly shows that there is a relationship between the number of daily hours of digital screen engagement and a mental well being score. Various forms of engagement (such as streaming, gaming and smartphones) had very similar patterns of relationships with mental health symptoms. This corroborates the evidence.

In conclusion, use of technology by students may negatively impact their mental health to some extent. Perhaps the influence is only negative when certain conditions are met and not all people are influenced by the use of technology equally, but converging evidence tells us that there is a tendency for excessive screen times to be correlated with some mental health problem, especially anxiety and depression.